Iteration 1

Start $m = [-1.45, 0.87]$ $b = 0.88$ $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$ $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

① $\hat{y} = Xm + b \cdot 1$

$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$

$= \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$

② $e = \hat{y} - y = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

$= \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$

③ $J = \frac{1}{2}(e_1^2 + e_2^2)$

$= \frac{1}{2}((-2.96)^2 + (-2.22)^2)$

$= 6.845$

④ Gradients: $\frac{dJ}{dm} = X^T e = [1(-2.96) + 4(-2.22), 3(-2.96) + 10(-2.22)]$

$= [-11.84, -31.08]$

$\frac{dJ}{db} = 1^T e = -2.96 + (-2.22)$

$= -5.18$

⑤ $\alpha = 0.01$

$m = m - \alpha \frac{dJ}{dm}$

$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$

$m = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$

$b = b - \alpha \frac{dJ}{db}$

$= 0.88 - 0.01(-5.18)$

$b = 0.9318$

∴ After updated step $m = [-1.3316, 1.1808]$ $b = [0.9318]$

For new cost;

$\hat{y} = Xm + b \cdot 1$

$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$

$= \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$

$= \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$

Scanned with CamScanner

$$e = \hat{y} - y = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$J = \frac{1}{2}(e_1^2 + e_2^2) = \frac{1}{2}((-1.8574)^2 + (1.4134)^2)$$

$$= 2.7238$$

∴ The new J is 2.7238